



AniClipart: Clipart Animation with Text-to-Video Priors

Ronghuan Wu¹ · Wanchao Su² · Kede Ma¹ · Jing Liao¹

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Abstract

Clipart, a pre-made graphic art form, offers a convenient and efficient way of illustrating visual content. Traditional workflows to convert static clipart images into motion sequences are laborious and time-consuming, involving numerous intricate steps like rigging, key animation and in-betweening. Recent advancements in text-to-video generation hold great potential in resolving this problem. Nevertheless, direct application of text-to-video generation models often struggles to retain the visual identity of clipart images or generate cartoon-style motions, resulting in unsatisfactory animation outcomes. In this paper, we introduce AniClipart, a system that transforms static clipart images into high-quality motion sequences guided by text-to-video priors. To generate cartoon-style and smooth motion, we first define Bézier curves over keypoints of the clipart image as a form of motion regularization. We then align the motion trajectories of the keypoints with the provided text prompt by optimizing the Video Score Distillation Sampling (VSDS) loss, which encodes adequate knowledge of natural motion within a pretrained text-to-video diffusion model. With a differentiable As-Rigid-As-Possible shape deformation algorithm, our method can be end-to-end optimized while maintaining deformation rigidity. Experimental results show that the proposed AniClipart consistently outperforms existing image-to-video generation models, in terms of text-video alignment, visual identity preservation, and motion consistency. Furthermore, we showcase the versatility of AniClipart by adapting it to generate a broader array of animation formats, such as layered animation, which allows topological changes.

Keywords Clipart animation · Text-to-video diffusion · Score distillation sampling · As-rigid-as-possible shape deformation

1 Introduction

Clipart, a collection of pre-made graphic elements, offers an efficient solution for quickly enhancing visual content without the need for custom artwork. Its user-friendly nature and diverse styles make it an invaluable resource for improving the visual appeal of documents, presentations, and websites. Animated clipart further enhances these benefits by captur-

ing user attention, facilitating the communication of critical messages, and adding an element of enjoyment to the content.

Traditionally, animating existing clipart has been a meticulous and labor-intensive process, involving steps such as rigging, keyframing, inbetweening, and precise control of timing and spacing. Recent advancements in Text-to-Image (T2I) synthesis, exemplified by Stable Diffusion (Rombach et al., 2022), have revolutionized the creation of high-quality clipart, shifting it from a manual to a more automated process. Despite these advancements, the automatic animation of clipart remains a relatively unexplored area.

The growing demand for animated clipart and the labor-intensive nature of its creation highlight the need for a system that can animate existing clipart with minimal or no manual intervention. A promising approach involves leveraging recent Text-to-Video (T2V) models, which can simultaneously accept a text prompt and an image (Xing et al., 2023a; Chen et al., 2023a, 2024; Zhang et al., 2023b). However, these models are inadequate for generating clipart animations for two main reasons. First, there is a significant discrepancy in motion patterns between videos produced

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✉ Jing Liao
jingliao@cityu.edu.hk

Ronghuan Wu
rh.wu@my.cityu.edu.hk

Wanchao Su
wanchao.su@monash.edu

Kede Ma
kede.ma@cityu.edu.hk

¹ City University of Hong Kong, Hong Kong, Hong Kong

² Monash University, Melbourne, Australia

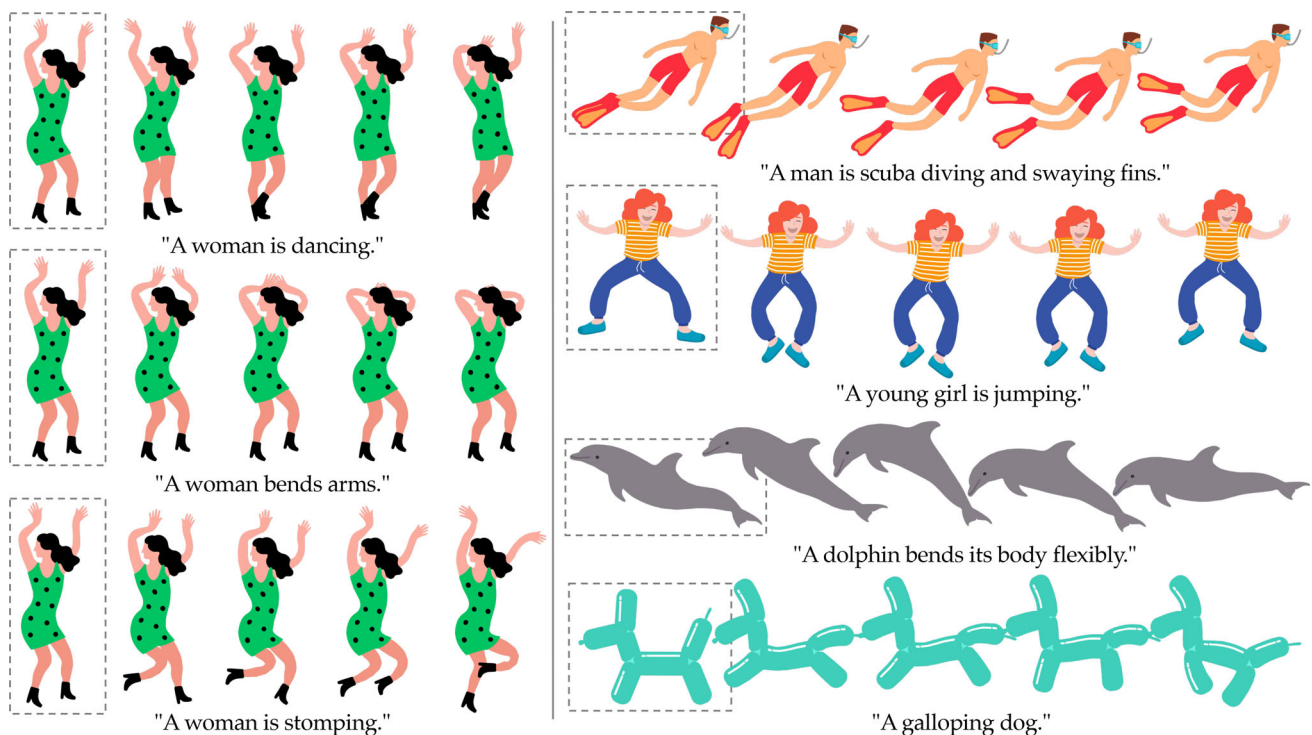


Fig. 1 Our AniClipart creates clipart animations guided by text prompts while preserving the visual identity of the input static clipart and achieving frame-to-frame consistency. The left panel displays different

animations generated from the same clipart input, guided by different text prompts. On the right panel, we present animations with diverse input categories. We denote the initial clipart with a dashed-line box

by existing large-scale T2V models and those typical of clipart animations. Training datasets for video generation models predominantly consist of natural-style videos that exhibit realism, complexity, and subtlety in motion. In contrast, cartoon-style clipart animations prioritize simplicity and clarity to convey messages efficiently. Second, video generation models often compromise the visual identity and distinct features of the original clipart due to the rich textures present in the output videos, such as flickering backgrounds and blurred patches. This disparity in motion paradigms and the presence of rich textures render state-of-the-art video generative models unsuitable for clipart animation.

In this paper, we introduce our system, *AniClipart*, which utilizes pretrained large-scale T2V models to animate clipart and align it with a text prompt, addressing the aforementioned challenges. Inspired by the standard animation pipeline, we define keypoints on the clipart and extract their trajectories from the T2V model to generate animations. By driving the clipart using these keypoints, we avoid direct pixel manipulation, thereby preserving the visual identity of the clipart. Additionally, by confining the motion to selected keypoints, we achieve shape-level animations that significantly reduce motion complexity and bridge the domain gap between different motion patterns.

Specifically, we assign Bézier curves as motion trajectories for each keypoint, determining their positions in each keyframe and ensuring smooth transitions between frames. To align the motion with a text prompt, we introduce Video Score Distillation Sampling (VSDD) loss to optimize the parameters of the Bézier trajectories, distilling motion knowledge from a pretrained T2V diffusion model. Score Distillation Sampling (SDS), a technique introduced in Dreamfusion (Poole et al., 2022), extends the knowledge acquired from pretrained diffusion models to tasks beyond their original scope. This technique is particularly valuable in fields with limited data availability, such as generating 3D models (Mildenhall et al., 2020) and vector graphics (Jain et al., 2022) from pretrained image diffusion models. To extend SDS from the image domain to the video domain, we input the animated clipart into a video diffusion model (Wang et al., 2023a) and assess the disparity between the current motion and the video diffusion model's motion prior based on the text description. This process enables us to achieve precise and contextually appropriate animation results.

A crucial aspect of clipart animation is preserving the visual identity of the input, which involves maintaining its local details and shape rigidity. To achieve this, in addition to the SDS loss, we propose incorporating a skeleton loss in the trajectory optimization process. This skeleton

loss constrains the length variation of the skeleton formed by keypoints, ensuring shape regularity. The optimization of both losses is integrated with a 2D shape deformation algorithm, specifically the As-Rigid-As-Possible (ARAP) shape manipulation (Igarashi et al., 2005), to preserve deformation rigidity. We further adapt the ARAP algorithm to be differentiable, allowing it to warp the clipart to a new pose defined by the updated keypoint positions while enabling backpropagation of gradients from the image-level loss to update the keypoints.

Extensive experiments and ablation studies demonstrate AniClipart’s ability to generate vivid and attractive clipart animations from text descriptions across various subjects, including humans, animals, and objects. Our system also supports layered animation to handle motion patterns involving topological changes and self-occlusion. Our contributions in this work can be summarized as:

1. We introduce a novel system capable of generating animations for clipart based on textual descriptions, marking a step forward in automatic animation generation.
2. We successfully utilize VS2S loss to distill motion from large-scale T2V models and apply it to optimize keypoint trajectories, resulting in semantically meaningful motions while creating abstract cartoon-style clipart motion patterns.
3. We integrate shape deformation algorithms with our skeleton loss to effectively maintain the visual identity of the original clipart across the generated animation.

2 Related Work

This section summarizes the prior works closely related to our approach. Given the breadth of topics related to 2D character animation, a comprehensive survey of each related topic is beyond the scope of this study. In this paper, we develop a system to animate existing 2D cartoon clipart based on text input (Sect. 2.1). Instead of training a feed-forward network from scratch, we leverage a pretrained video generation model (Sect. 2.2) and employ score distillation sampling loss (Sect. 2.3) to extract motion.

2.1 2D Cartoon Animation

Traditional animation production is a time-consuming and labor-intensive process. Figure 2 illustrates a simplified pipeline for 2D cartoon animation, emphasizing three crucial stages: rigging, keyframing, and inbetweening. Recent research has focused on creating automated algorithms to streamline these essential processes.

Rigging is a foundational step in animation that constructs skeletons for characters. Along with skinning algo-

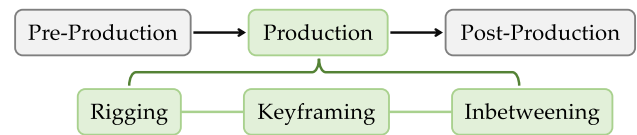


Fig. 2 A simplified animation pipeline

rithms (Forstmann & Ohya, 2006; Kavan et al., 2007; Le and Lewis, 2019), transformations applied to the skeletons can be distributed across the entire shape, alleviating the need for animators to draw every frame manually. Automatic rigging techniques can be classified into two categories: (1) The first category maps a predetermined skeletal template onto the character (Baran & Popovic, 2007; Li et al., 2021), but it lacks flexibility when characters do not fit the template. (2) The second category can generate rigs for broader categories. Traditional approaches (Au et al., 2008; Huang et al., 2013; Tagliasacchi et al., 2012) analyze geometric features of the mesh to produce curve-skeletons but often ignore movable parts. Recently, deep learning-based methods (Liu et al., 2019; Xu et al., 2020) demonstrate a robust capacity for rig generation, offering a viable alternative for animators.

Keyframing involves defining character poses in keyframes. Animators create a deformable puppet, typically represented by a triangular mesh, on the characters and manually adjust handles (e.g., skeletons and control points generated in rigging) across the keyframes. However, defining key poses is still challenging for those without animation skills (Fan et al., 2018). Researchers have thus focused on developing methods to transfer motion to predefined puppets. Hornung et al. (2007) fitted a 3D model onto 2D characters and then animated them with 3D motion. Animated Drawings (Smith et al., 2023) creates rigged characters from children’s art, and maps predefined human motion for animation. Bregler et al. (2002) and DeJuan & Bodenheimer (2006) captured and transferred motion from existing cartoons, enriching the expressiveness of animations with diverse movements. Furthermore, recent advancements include extracting motion from videos. Live Sketch (Su et al., 2018) transfers video-derived motion trajectories to sketches, Pose2Pose (Willett et al., 2020) uses cluster algorithms to select key poses from video-tracked performances, and AnaMoDiff (Tanveer et al., 2024) warps the character according to the optical flow extracted from a driving video.

Inbetweening involves inserting drawings between keyframes to transform choppy and disjointed animations into smooth and seamless sequences. A straightforward approach is interpolating between keyframes with shape interpolation algorithms (Alexa et al., 2000; Baxter et al., 2008; Baxter et al., 2009; Whited et al., 2010; Fukusato & Maejima, 2022; Fukusato & Morishima, 2016; Kaji et al., 2012; Chen et al., 2013). Nevertheless, these algorithms often struggle to produce life-like movements (Fukusato

et al., 2023). Recognizing animation inbetweening as a subset of frame interpolation opens the door to applying recent advancements in video frame interpolation techniques (Liu et al., 2017; Niklaus et al., 2017a, b; Jiang et al., 2018; Niklaus & Liu, 2018; Xu et al., 2019; Niklaus & Liu, 2020; Park et al., 2020; Sim et al., 2021; Huang et al., 2022; Reda et al., 2022; Lu et al., 2022). However, Siyao et al. (2021) pointed out that animation videos possess their distinct features, such as flat color regions and exaggerated motions, which lead general video interpolation models to perform poorly. Consequently, there is a growing focus on developing algorithms tailored for the animation domain (Siyao et al., 2021; Li et al., 2021; Chen & Zwicker, 2022; Xing et al., 2024).

In animation production, characters can be stored in both bitmap and vector graphics formats, allowing animators to select the most suitable format for their specific requirements. While both formats share a similar animation pipeline, vector graphics animation requires extra effort. For example, researchers have developed efficient representation learning algorithms (Carlier et al., 2020; Cao et al., 2023), improved data structures to achieve topological changes (Dalstein et al., 2015), specialized skinning algorithms (Liu et al., 2014), and advanced inbetweening techniques (Siyao et al., 2023). These advancements significantly facilitate animators in creating vector animations.

In our paper, we develop a system that automates these three crucial steps, supporting animation for both bitmap and vector graphics characters. We use a hybrid rigging algorithm to create accurate skeletons for humans and objects. Keyframes are selected along motion trajectories and refined using video diffusion models. These motion paths allow us to interpolate frames at any point, ensuring smooth animation.

2.2 Video Generation

Generating videos from text descriptions is challenging due to the scarcity of high-quality text-video pairs, the variable lengths of videos, and the high computational cost involved. As T2V generation extends the principles of T2I creation, the development of T2V models follows a trajectory similar to that of T2I: transformers (Villegas et al., 2022; Wu et al., 2022; Hong et al., 2022; Kondratyuk et al., 2023) and diffusion models (Ho et al., 2022; Gupta et al., 2023). Regarding training strategy, one research direction involves training a T2V model from scratch utilizing both image and video data to learn their joint distribution (Singer et al., 2022; Ho et al., 2022). Such a method demands significant training time and computational resources. Recent studies have explored the integration of additional temporal consistency modules into existing T2I models (*e.g.*, Stable Diffusion (Rombach et al., 2022)), using text-video data pairs to train the newly-added layers (or finetune the entire model) (Singer et al., 2022; Blattmann et al., 2023b; Girdhar et al., 2023; Ge et al., 2023;

Yuan et al., 2024; Guo et al., 2023b). This training paradigm leverages the robust image prior of T2I models to maintain spatial consistency, while additional dedicated modules learn to extend temporal dimension, thereby accelerating convergence. Some work introduces additional image conditions to these T2V models, enabling them to perform Image-to-Video (I2V) synthesis (Blattmann et al., 2023a; Gu et al., 2023; Wang et al., 2023b; Xing et al., 2023a; Zhang et al., 2023b). This is typically achieved by concatenating two input images with noisy frame latents (Xing et al., 2023a), or by adding an extra image encoder (Guo et al., 2023a) that is similar to the ControlNet (Zhang et al., 2023a). However, attempting to animate clipart using these models often leads to pixel-level artifacts in the resulting videos, such as distortion and blurriness. Moreover, since these models are primarily trained on natural videos, they are unsuited for generating clipart animation. One way to overcome these limitations is to bypass the direct generation of video content. The recent approach introduced by Li et al. (2024) effectively synthesizes natural and oscillatory motions, such as those seen in trees and flowers, by employing a diffusion model that operates in the frequency domain—a more fundamental and lower-dimensional representation. However, the repetitive local motion patterns typically observed in natural videos significantly differ from those in clipart animations. We aim to create animations that exhibit more comprehensive motion and are closely aligned with high-level semantics provided by textual prompts. Consequently, the framework suggested by Li et al. (2024) may not be fully compatible with our objectives. We propose a solution that involves distilling motion from T2V diffusion models and leveraging it to optimize the motion trajectories of keypoints defined on clipart. By adopting this approach, we can better regulate the motion and achieve the desired cartoon animation style required by clipart.

2.3 Score Distillation Sampling

T2I diffusion models are pretrained on an extensive collection of text-image pairs, providing a comprehensive understanding of relations between text descriptions and 2D images. Utilizing image diffusion models beyond image synthesis and transferring prior knowledge to new tasks, particularly those with limited annotated data, have become a prominent research area. DreamFusion (Poole et al., 2022) has shown the feasibility of generating 3D representations (*e.g.*, NeRF (Mildenhall et al., 2020)) from a text description using only 2D image diffusion models, with a newly devised loss function—score distillation sampling loss. SDS is similar to the diffusion model training loss but distinctively excludes the U-Net Jacobian term. This modification eliminates the need for computationally expensive backpropagation within the diffusion model and guides the optimization process toward aligning the images with the textual conditions.

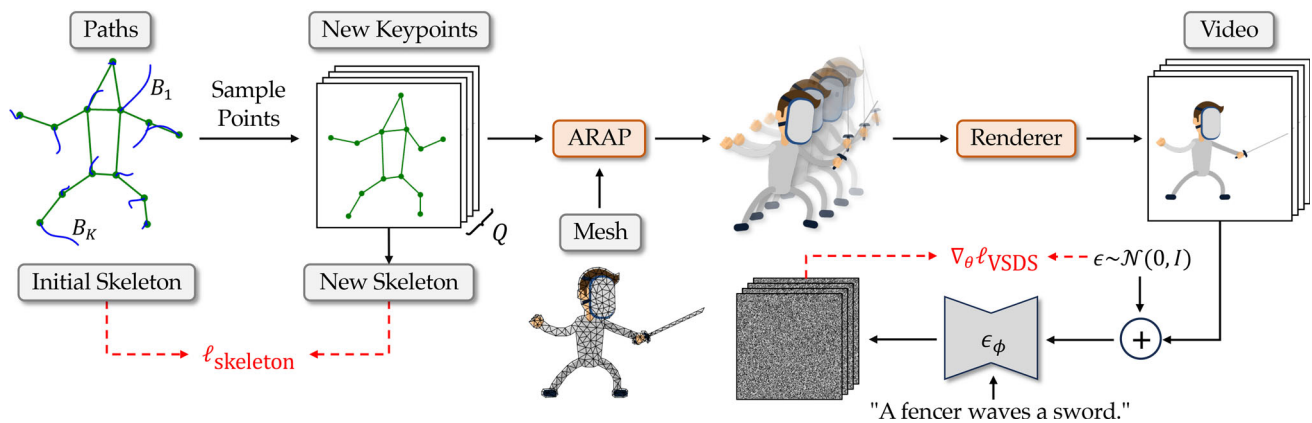


Fig. 3 The system diagram of AniClipart. There are K Bézier paths, $\{B_i\}_{i=1}^K$, serving as the motion trajectories of K keypoints. We select frame numbers Q , define timesteps, and sample points along the adjusted Bézier paths to determine new positions of keypoints. These

new keypoints guide the ARAP algorithm to warp the triangular mesh, transforming the clipart to new poses. Subsequently, we use a differentiable renderer to convert the updated shapes into a video V and send it to a video diffusion model ϵ_ϕ to compute loss functions

Subsequently, SDS loss is widely used to optimize other generation tasks, including artistic typography (Iluz et al., 2023; Tanveer et al., 2023), vector graphics (Jain et al., 2022), sketches (Xing et al., 2023b; Qu et al., 2023), meshes (Chen et al., 2023b) and texture maps (Metzger et al., 2023; Tsalicoglou et al., 2023). The advent of T2V diffusion models (Ni et al., 2023; Wang et al., 2023b; Dai et al., 2023; Chen et al., 2023a, 2024; Bar-Tal et al., 2024) naturally projects the image SDS loss to extended dimensional applications. Recently, video SDS loss has been applied to create vector sketch animations (Gal et al., 2023). This research involves predicting the positions of all points within a sketch for each frame and using VSDS loss to align the entire animation with a textual prompt. However, this technique falls short for clipart animation due to its inability to maintain shape rigidity and the resultant inconsistency in animation across frames. In our study, we apply VSDS loss specifically to refine the motion trajectories of keypoints.

3 AniClipart

In this section, we provide a detailed elaboration of our AniClipart. We begin with an overview of our method (Sect. 3.1), followed by the introduction of the implementation in three main aspects: clipart preprocessing (Sect. 3.2), Bézier-driven animation (Sect. 3.3) and loss functions (Sect. 3.4).

3.1 Method Overview

In the clipart preprocessing stage, we detect keypoints, build skeletons, and construct a triangular mesh over the clipart to facilitate shape manipulation. In the subsequent Bézier-driven animation phase, we attach a Bézier curve to each of



Fig. 4 Clipart preprocessing. We use UniPose (Yang et al., 2023b) to detect keypoints and skeletons of articulated objects (e.g., humans)

these keypoints, serving as the motion trajectories regulating these keypoints over time. When the keypoints reach their new positions at a specific time, the differentiable as-rigid-as-possible shape deformation algorithm (Igarashi et al., 2005) adjusts the entire clipart to conform to these new positions. Such continuous deformation results in a series of consecutive frames, which can be naturally concatenated to create a video V . This video is then sent to a pretrained video diffusion model (Wang et al., 2023a) to compute the video score distillation sampling loss (Poole et al., 2022). Along with a skeleton length loss to maintain the clipart's shape integrity, we optimize the parameters of Bézier curves to ensure that the animation aligns with the provided text description. Figure 3 illustrates the overall workflow of our method, with detailed explanations for each component provided in the following subsections.

3.2 Clipart Preprocessing

The initial step in our animation procedure, akin to traditional animation, involves character rigging, where we detect keypoints on the clipart and construct skeletons between these points. State-of-the-art keypoint detection algorithms (Sun et al., 2023; Ye et al., 2022; Mathis et al., 2018; Ng et al.,

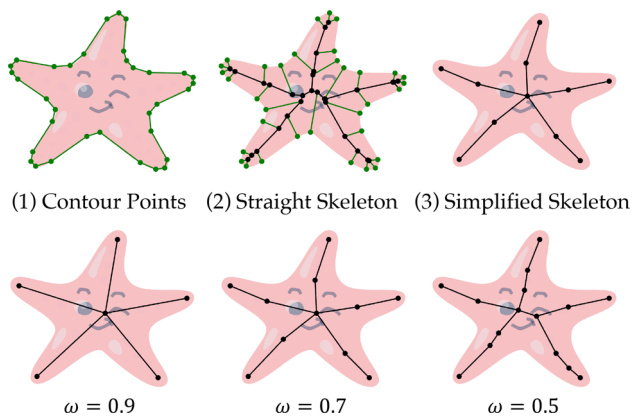


Fig. 5 The first row shows the keypoint detection procedure for broader categories. The second row shows the effect of different choices of ω

2022; Xu et al., 2022; Jiang et al., 2023; Yang et al., 2023a) excel at assigning predefined keypoints to articulated characters. However, these methods are often limited to object categories within their training datasets, posing a challenge when applied to clipart, which includes a diverse range of cartoon objects.

To address the limitations of existing keypoint detection methods, we adopt a hybrid approach. We utilize the UniPose algorithm (Yang et al., 2023b) to predict keypoints and construct skeletons with semantic meanings for articulated objects, such as humans (see Fig. 4). UniPose integrates keypoint detection tasks into a comprehensive, end-to-end, prompt-driven framework capable of identifying keypoints across a wide range of objects, including articulated, rigid, and soft objects. We choose UniPose for its generalizable and effective performance on cartoon-style images, and its extendability for adding extra keypoints. For broader categories, like sea animals and plants, our keypoint detection algorithm involves three steps (see Fig. 5 for illustrations):

1. *Binary Conversion and Boundary Detection.* We convert colorful clipart into binary (*i.e.*, black-and-white) images, with characters rendered in black and the background in white. Using OpenCV's (Bradski, 2000) `findContours` function, we detect points along the characters' boundaries. These points are then connected counter-clockwise to construct the boundary edges;
2. *Skeleton Generation.* We generate straight skeletons by inwardly propagating edges in their perpendicular directions. All edges move at the same speed, and when they collide, the intersected positions are marked as keypoints (Cacciola, 2004). This can be done using an existing library such as `scikit-geometry`.
3. *Skeleton Pruning and Simplification.* The straight skeleton structure often contains excessive details for animation. We prune these unnecessary outer skeletons

(illustrated by green line segments in Fig. 5) and merge adjacent segments if the distance between them is shorter than a predetermined threshold. Typically, we set the threshold as $\omega \cdot \frac{1}{N_S} \sum_{i=1}^{N_S} \text{length}(s_i)$, where $\omega = 0.7$ is the weight controlling the threshold, s_i is the i -th skeleton and N_S is the total number of skeletons. Skeleton simplification continues in several steps until no further merging occurs. Adjusting ω allows for the generation of skeletons with varying levels of complexity. In the second row of Fig. 5, we show skeletons of different complexities based on various ω choices. Empirically, $\omega = 0.7$ creates a reasonable skeleton structure for most categories.

After keypoint detection, we use a triangulation algorithm (Shewchuk, 1996) to create a mesh for deformation-based animation (see Fig. 4).

3.3 Bézier-Driven Animation

Prior research (Gal et al., 2023) forecasts new positions for sketch control points at distinct timesteps. This approach, however, occasionally struggles to ensure temporal coherence throughout the animation frames. Additionally, as the complexity of the sketch escalates, the technique often falls short of preserving the fidelity of the original design. These inconsistencies produce significant distortion for clipart animation. Users expect animations to flow smoothly, and any shape distortion could jeopardize the overall quality of the animation.

To address the problems above, we define motion trajectories on a few keypoints rather than control all the points. We employ 2D deformation algorithms to animate the entire clipart based on keypoint motion, reducing the search space for motion creation while maintaining the clipart's appearance. We particularly choose cubic Bézier curves as the motion trajectories due to their extensive application in design. These parametric Bézier curves can be flexibly manipulated by altering the coordinates of control points. We assign a cubic Bézier curve for each keypoint, with each curve's start point accurately aligned with the keypoint coordinate, ensuring the animation originates from the clipart's initial pose, as shown Fig. 6. The remaining three points of each curve (*i.e.*, two control points and an endpoint) are randomly initialized, with the initial curve length kept short to ensure moderate initial motion.

Formally, a clipart S contains K keypoints, denoted as $\{k_i\}_{i=1}^K$, corresponding to K cubic Bézier curves, $\{B_i\}_{i=1}^K$, where each curve B_i includes four control points, $B_i = \{p_i^j\}_{j=1}^4$. To monitor the progression of these keypoints over time, we define a sequence of timesteps, $T = \{t^q\}_{q=1}^Q$, with $t^q \in [0, 1]$ and Q denotes the total number of frames in the animation. At every timestep t^q , we sample points on each

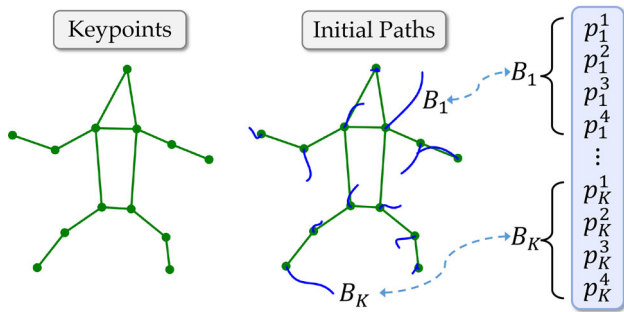


Fig. 6 Motion path initialization. We create a cubic Bézier curve for each keypoint, with the curve’s start point matching the initial keypoint position precisely. This means the animation begins with the object’s original position. The initial curves are short in our experiments, but we intentionally elongate them in this figure for clearer visualization

cubic Bézier curve to determine the new positions of the keypoints, resulting in a set of updated positions, $\{k_i^q\}_{i=1}^K$. These new positions allow us to adjust the triangular mesh grid using the ARAP deformation algorithm (Igarashi et al., 2005). ARAP enables users to deform 2D shapes intuitively by dragging several keypoints. This method employs a two-step closed-form algorithm that first determines the rotation of each triangle and then adjusts the scale. ARAP performs in real-time while minimizing distortion across the triangular mesh. Our approach makes the ARAP algorithm differentiable, allowing for the backpropagation of gradients. Subsequently, we warp the clipart, producing a new shape S^q . We then use a differentiable renderer $\mathcal{R}$ to convert this altered shape into a pixel image, denoted as $I^q = \mathcal{R}(S^q)$. Repeating this procedure across all defined timesteps gives us a sequence of frames, $I = \{I^q\}_{q=1}^Q$. These frames are then temporally stacked to create a video V for further processing.

In our approach, we arrange timesteps to create *looping animations*, inspired by the observation that clipart animation often exhibits periodic movement patterns. These animations are designed to repeat continuously without noticeable breaks at the end of the sequence. For a predetermined frame count Q , we begin by organizing timesteps in a monotonically increasing order but cap the sequence at $Q/2$ frames. This yields $\{t^q\}_{q=1}^{Q/2}$, where $t^1 = 0$, $t^{Q/2} = 1$, and $t^1 < t^2 < \dots < t^{Q/2}$. Afterward, to achieve a seamless looping effect, we invert this sequence and append it to the original, thereby generating a complete loop:

$$T_{\text{loop}} = \{t^0, \dots, t^{Q/2}, t^{Q/2}, \dots, t^0\}. \quad (1)$$

For our clipart examples, T_{loop} successfully creates animations with natural and rhythmic motions. For animations that do not require looping, designers are free to choose $T_{\text{no-loop}} = \{t^q\}_{q=1}^Q$, where $t^1 = 0$, $t^Q = 1$, and $t^1 < t^2 < \dots < t^Q$, allowing for flexibility in animation styles.

3.4 Loss Functions

We use video SDS loss to animate clipart based on user-provided text input. To maintain the structural integrity of the original clipart, we incorporate a skeleton preservation loss into the total loss function.

3.4.1 Video SDS Loss

Unlike the image SDS loss, which focuses on augmenting a 2D image with additional content based on a single image input, the VSDS loss takes a video as input. It guides content addition across a sequence of images while maintaining frame-to-frame consistency. In our approach, the updated motion trajectories are used to create a video $V = \{I^q\}_{q=1}^Q \in \mathbb{R}^{h \times w \times Q}$. We select an intermediate diffusion timestep t and infuse the video V with a randomly sampled noise ϵ , which results in $V_t = \alpha_t V + \sigma_t \epsilon$, where α_t and σ_t are parameters from the noising schedule. Following this, we pass the noise-injected video V_t and the text condition c to a pretrained T2V diffusion model (Wang et al., 2023a). The model will output a predicted noise $\epsilon_\phi(V_t, t, c)$. Then, we compute the VSDS loss as follows:

$$\nabla_\theta \ell_{\text{VSDS}} = \mathbb{E}_{t, \epsilon} \left[w(t) \left(\epsilon_\phi(V_t, t, c) - \epsilon \right) \frac{\partial V}{\partial \theta} \right], \quad (2)$$

where $w(t)$ is a constant that relies on α_t and θ is the control points of cubic Bézier curves. This loss measures the discrepancy between the model’s predicted noise and the actual noise ϵ , steering the Bézier-driven animation towards alignment with the text description. The optimization process updates the shape of Bézier curves gradually, and is conducted iteratively until it reaches convergence.

3.4.2 Skeleton Loss

Optimizing motion trajectories using $\nabla_\theta \ell_{\text{VSDS}}$ produces animations that align with the text input. Nevertheless, solely using VSDS for optimization struggles to preserve the clipart’s authentic appearance in the animation. To preserve the structure of the objects, we leverage the skeleton extracted by the keypoint detection algorithms (Sect. 3.2) to guarantee minimal deviation in skeletal component lengths from their original measurements. Suppose there are N_S skeletal components, $\{s_i\}_{i=1}^{N_S}$, in the original structure, we can get a new skeleton, $\{s_i^q\}_{i=1}^{N_S}$, at a specific frame q based on the updated keypoint positions $\{k_i^q\}_{i=1}^K$. The overall skeleton loss is:

$$\ell_{\text{skeleton}} = \frac{1}{N_S} \sum_{q=1}^Q \sum_{i=1}^{N_S} (\text{length}(s_i^q) - \text{length}(s_i))^2. \quad (3)$$

3.4.3 Total Loss and Optimization

The weighted average of two terms then defines the final loss function:

$$\ell_{\text{total}} = \nabla_{\theta} \ell_{\text{VSDS}} + \lambda \ell_{\text{skeleton}}, \quad (4)$$

where $\lambda = 25$ acts as a weighting factor, balancing the importance of text alignment with the preservation of appearance. Given that the ARAP (Igarashi et al., 2005) is a closed-form algorithm that solves equations to determine new positions of vertices on triangular meshes, our entire process is differentiable, and the gradient can be backpropagated to update the motion trajectories.

4 Experiments

To evaluate the effectiveness of our proposed *AniClipart*, we conducted extensive quantitative and qualitative experiments, demonstrating our method’s superiority over alternative solutions. This section is organized as follows: first, we introduce the experimental setup of our system (Sect. 4.1). Next, we discuss the evaluation metrics (Sect. 4.2). We then present a comparison between our model and a closely related study (Gal et al., 2023), as well as leading T2V generation models, highlighting the distinctions and contributions of our approach within the clipart field (Sect. 4.3). Furthermore, we present the outcomes of ablation studies to validate our method’s superiority over alternative techniques in the animation procedure (Sect. 4.4). Finally, we demonstrate our method’s capability to handle more challenging topology-changing cases by incorporating layer information (Sect. 4.5).

4.1 Experiment Setup

We collected 30 cliparts from Freepik,¹ including 10 humans, 10 animals, and 10 objects, with each clipart resized into 256×256 . We optimized the positions of control points for Bézier curves with an Adam (Kingma & Ba, 2014) optimizer for 500 steps, with a learning rate of 0.5. We used ModelScope text-to-video (Wang et al., 2023a) as the diffusion backbone, setting the entire animation to 24 frames and adjusting the SDS guidance scale to 50. The entire process of one clipart animation, conducted on an NVIDIA RTX A6000, requires about 25 minutes with 26GB memory consumption.

Clipart is commonly stored and distributed in two formats: bitmap and vector graphics. In our experiments, we concentrated on clipart in the Scalable Vector Graphics (SVG) format, which offers enhanced visual quality and more varied

outcomes. The scalability of SVG allows for the production of high-resolution frames, while its layered structure supports the creation of complex animations with topological changes. Each SVG file includes multiple geometric shapes known as *Paths*, comprising sequences of control points. During the animation, we update the control points based on the barycentric coordinates after adjusting the triangular shape with the ARAP algorithm. We then use DiffVG (Li et al., 2020) to convert clipart from vector graphics format to bitmap, which is subject to video diffusion models. Our method is not limited to vector clipart; rasterized clipart can be processed using the same procedure with comparable output quality. The primary distinction lies in the warping operation for bitmap images, wherein pixel patches (instead of control points) are relocated to new positions.

4.2 Evaluation Metrics

To evaluate the quality of AniClipart, we conducted assessments at both the bitmap-level and vector-level of the generated animations.

4.2.1 Bitmap-Level Metrics

To compare our system with state-of-the-art methods, we focus on two objectives for evaluating the quality of the generated animation: (1) maintaining the visual identity of the input clipart, and (2) ensuring alignment between the text descriptions and the resulting animations. For visual identity preservation, we employed CLIP (Radford et al., 2021) to calculate the average cosine similarity scores between the feature representations of the input clipart image and each frame of the generated animation. To measure the alignment between the videos and text descriptions, we utilized X-CLIP (Ni et al., 2022), an extension of the CLIP model adapted for the video domain, to compute similarity scores.

4.2.2 Vector-Level Metrics

Consistent with prior research (Gal et al., 2023), we observed that CLIP and X-CLIP fail to capture the nuanced differences between animations produced in the ablation study. Consequently, we leveraged vector graphics outputs as a more precise means of evaluation. To be more specific, we computed three distinct metrics:

1. *Dynamism* This metric measures the motion magnitude in the results by calculating the average trajectory length of all keypoints in an animation. When the keypoints’ locations are predicted directly for each frame (see “w/o Trajectory” in Fig. 10), we connect two keypoints in consecutive frames as the pseudo-trajectory.

¹ <https://www.freepik.com/>

2. **Temporal Consistency** A vector graphic's shape is defined by a set of control points along the contours. We calculate the Hausdorff distance (represented by D_H) between two sets of control points in adjacent frames, using $1 - D_H$ as the similarity score. We average these scores throughout the animation sequence to determine the level of consistency.
3. **Distortion** We first calculate the curvature of each point in the initial clipart. For each frame, we recalculate the curvature, and the difference between the initial curvature and the curvature for each frame determines the level of distortion.

4.3 Comparison to State-of-the-Art Methods

Given the scarcity of existing works on direct clipart animation, we compared our system to two alternatives: (1) a prior work (Gal et al., 2023) that concentrates on animating vector sketches, and (2) the state-of-the-art T2V generative models.

AniClipart versus Gal et al. (2023). Both our AniClipart and Gal et al.'s method aim to extract motion priors using pretrained T2V diffusion models with VSDS losses. Despite this commonality, there are two major differences. Firstly, Gal et al.'s method lacks a rigidity constraint, leading to noticeable alterations in the character's appearance across different frames, as shown in Fig. 7. While minor distortions are acceptable in abstract sketches, they become catastrophic for clipart with delicate parametric edges during animation. Gal et al.'s algorithm drastically modifies the appearance of all characters, rendering it unsuitable for clipart applications. In contrast, our method incorporates the ARAP deformation algorithm to explicitly maintain the rigidity of the clipart, ensuring a consistent appearance throughout all frames and achieving the highest CLIP similarity scores, as shown in Table 1.

Another key distinction lies in the optimization target. Gal et al.'s method predicts new positions for *all* control points in a sketch, whereas ours focuses on altering only a small number of detected keypoints. Our method refines a motion trajectory for each keypoint, ensuring smooth motion regardless of the complexity or the quantity of control points in the original clipart. The number of keypoints is orders of magnitude less than that of the control points in the entire sketch (ours 13 versus Gal et al.'s 1067 for the fencer example), effectively reducing the searching space of the optimization process. Our keypoint-focused approach with the ARAP effectively maintains the integrity of the *in-frame* appearance and produces semantically consistent motion across the whole clipart. Additionally, predicting new positions for control points (or keypoints) can sometimes lead to inconsistent movements across consecutive time steps (see the supplementary video). In contrast, our method of distributing timesteps along a Bézier curve (Sect. 3.3) successfully

eliminates the above issue, ensuring an *inter-frame* motion consistency.

AniClipart versus T2V Models. In our experiments, we rasterized vector clipart into bitmap images and incorporated ModelScope (Wang et al., 2023a), VideoCrafter2 (Chen et al., 2024), DynamiCrafter (Xing et al., 2023a), I2VGen-XL (Zhang et al., 2023b) and ToonCrafter (Xing et al., 2024) to generate respective videos. In particular, ModelScope, the video backbone model of our method, cannot accept text descriptions and images simultaneously. Therefore, we mixed the initial image with random noise and sent the noise-injected image to ModelScope to create videos. We set the blending ratio at 0.84. A higher ratio can create larger motions but at the expense of losing visual identity, while a lower ratio can preserve the visual identity better but create almost static objects. ToonCrafter is a T2V model fine-tuned on an anime dataset with interpolation capability. It requires both the first and last frames as input. We use the initial clipart as the beginning frame. To avoid manually creating the last frame, we automatically generate it using IP-Adapter (Ye et al., 2023), which creates variations of the input object with consistent style and content.

In Fig. 8, we show the representative frames in the videos created by T2V models. We observe two primary issues: (1) T2V models often generate some pixel-level artifacts, including shape distortion and blurred patches, thus cannot well preserve the original character's visual identity, leading to lower identity preservation scores compared to ours. (2) The motion generated by T2V models is less pronounced compared to that of our system. Specifically, ModelScope strikes a decent balance between identity preservation and motion generation with a carefully chosen blending ratio but sometimes fails to drive the character to move at all, resulting in an inferior score in text-video alignment (Table 1). VideoCrafter2 may produce reasonable videos for a few clipart images. But in many cases, it fails to add motion to the objects. DynamiCrafter tends to add extra shapes and textures to the input clipart without introducing noticeable motions, thereby having the lowest text-video alignment score in Table 1. In addition, I2VGen-XL can produce moderate animations (*e.g.*, the fencer) but at times drastically changes the appearance of the subjects (*e.g.*, the dancing woman). ToonCrafter may be a suitable option for creating animations with a large motion range. However, despite maintaining the style, the final frame generated by IP-Adapter still fails to accurately preserve the characters' identity. Additionally, the interpolated animation produced by ToonCrafter resembles linear interpolation, lacking the trajectory complexity offered by our method. Consequently, the text-video alignment scores of these models do not match up to the standards set by our methods.

User Study. We conducted a subjective user study to evaluate the perceptual quality of our AniClipart over other

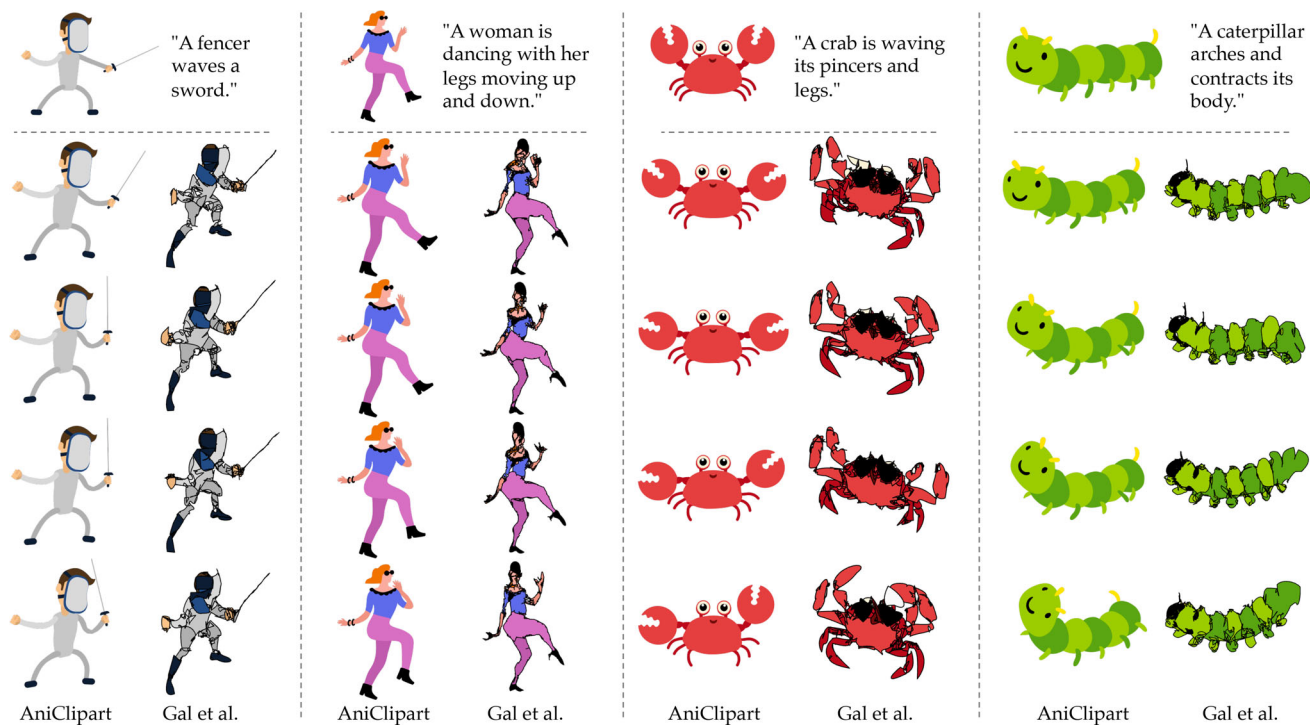


Fig. 7 AniClipart versus Gal et al. (2023). We select four consecutive frames from the output animation to illustrate the differences. Our method effectively maintains the original shape, producing reasonable

and visually pleasing animations. On the other hand, the approach by Gal et al. drastically alters the characters' appearance and lacks continuity and consistency across frames

Table 1 Quantitative comparisons to Gal et al. (2023) and state-of-the-art T2V models

Method	Identity Preservation $\uparrow$	Text-Video Alignment $\uparrow$
Ours	0.9414	0.2071
Gal et al	0.8379	0.1879
ModelScope	0.8618	0.2021
VideoCrafter2	0.8403	0.1992
DynamiCrafter	0.8008	0.1741
I2VGen-XL	0.8813	0.1999
ToonCrafter	0.9292	0.2003

The [bold] in the tables represents the best performance

comparing methods. The user study comprised two distinct tasks to visually evaluate (1) the preservation of visual identity and (2) the alignment between text and video. In the first task, we presented 30 static cliparts to participants, with each clipart followed by seven animations created by AniClipart, Gal et al., ModelScope, VideoCrafter2, DynamiCrafter, I2VGen-XL, and ToonCrafter, respectively. Each participant was asked to rate the agreement of good quality in the generated animations in terms of identity preservation. We employed the five-point Likert scale on each criterion, with 1 representing *Strongly Disagree* and 5 representing *Strongly Agree*. In the second task, we presented the participants with 30 textual prompts, each followed by the same set of seven animations as in the first task. Participants were

then asked to rate the agreement of well-alignment of each animation to the text prompt.

We conducted the user study via an online questionnaire and received complete feedback from 42 participants. To analyze the results, we calculated the average quality score of each animation across the 30 examples for both tasks, with the results presented in Fig. 9. We also conducted one-way ANOVA tests, and the p -values of the two tasks were both less than 0.001. The statistical results show that AniClipart significantly outperforms other methods. In terms of visual identity preservation, our system's deformation algorithm ensures that the characters' details are accurately preserved. Regarding text-video alignment, Gal et al. (2023) and ToonCrafter can sometimes produce ani-

mations that closely match the text prompt, but the identity is greatly degraded. ModelScope occasionally produces reasonable animations that maintain identity, benefiting from a carefully optimized blending ratio. However, the other three T2V models often fail to produce noticeable motions that align with the text, revealing a limitation in their ability to ensure text-video coherence.

4.4 Ablation Study

In this section, we elaborate on the ablation study conducted to validate the effect of each essential component of our system, with the quantitative results detailed in Table 2. We also show a qualitative comparison of the ablation study in Fig. 10. In the following of this subsection, we describe each component in detail.

4.4.1 ARAP Deformation

To analyze the effectiveness of ARAP incorporation, we substituted the deformation module with Linear Blend Skinning (LBS), another prevalent algorithm in shape deformation, where the new position of each point is determined by the spatially weighted transformations of the keypoints. In our study, we implemented LBS with bounded biharmonic weights (Jacobson et al., 2011). Nonetheless, LBS frequently produces animations with unrealistic deformations, such as the shoulder squeezing in the dancing man case and leg distortion in the dog case in Fig. 10. In Table 2, LBS also has a higher distortion score when compared to ours. For settings without any 2D deformation algorithm employed, the generation aligns with the settings of the comparison between ours and Gal et al. (2023) (see Fig. 7), which is inferior to our method.

4.4.2 Bézier-Driven Animation

We define Bézier curves as the motion trajectories for keypoints, aiming to achieve smooth and fluid animations. To understand their significance, we eliminated all motion paths and predicted the keypoints' new positions in each frame. This setting could create animations matching the text description, but unfortunately, it failed to preserve frame-to-frame consistency, resulting in noticeable flickering effects ("w/o Trajectory" entries and Fig. 10). Its dynamism score is the highest in Table 2 since the positions of keypoints change drastically in each frame, leading to the lowest frame consistency score. We further linked the keypoints in two frames to visualize the pseudo-trajectories in Fig. 10, validating the significant inconsistency across frames. When we replaced Bézier curves with straight lines, the generated motions were overly simple. We refer to this setting as "Linear Interp." in the quantitative and qualitative results.

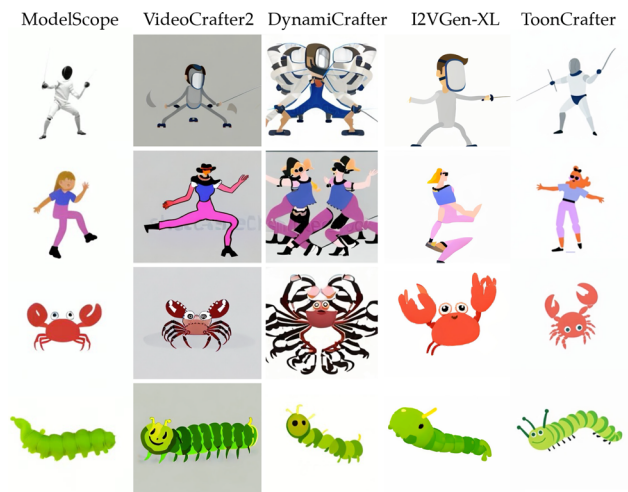


Fig. 8 Results of T2V models. These methods only preserve the semantics of the input clipart but fail to preserve the details, leading to low identity preservation scores. Furthermore, these approaches frequently yield animations with limited movement, resulting in inferior text-video alignment



Fig. 9 Subjective user study results

Notably, such straight-line guided results presented the highest consistency score and the lowest distortion rate in Table 2; we attribute such phenomenon to their limited motions, with little shape changes in the results. For instance, Fig. 13 displays the optimized trajectories with only two control points (*i.e.*, linear interpolation), where the motion shows limited range and the appearance shows little change. Please find the side-by-side comparison in the accompanying video to better compare the motions produced in the animation results. These two alternative comparison methods clearly show that cubic Bézier curves are important in creating realistic and lifelike movements.

4.4.3 Loss Functions

The video SDS loss $\nabla_{\theta} \ell_{VSDS}$ is crucial for generating motion trajectories that are in harmony with motion priors from text-to-video diffusion models. Replacing VSDS with another

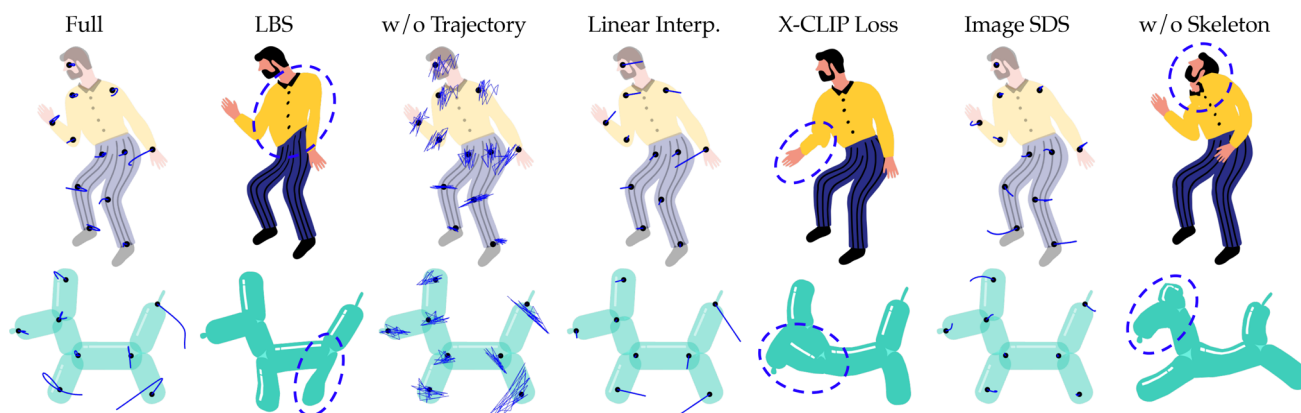


Fig. 10 Results of ablation study. Eliminating key components from our system could lead to animations with restricted movements (e.g., Linear Interp. and Image SDS), shape distortions (e.g., LBS, X-CLIP

loss and w/o Skeleton), and inconsistencies across frames (e.g., w/o Trajectory). Motion details are rendered by the overlaid blue trajectories, and shape details are highlighted by dash-lined circles

temporal-aware loss function, such as X-CLIP loss (Ni et al., 2022), does not produce reasonable motions aligned with the text prompt. Instead, the resulting animation displays unusual movements and significant distortions. Additionally, switching to an image-based SDS loss $\nabla_{\theta} \ell_{\text{SDS}}$ compromised the animation's relevance to the text input (the lowest dynamism score in Table 2). Furthermore, omitting skeleton loss ℓ_{skeleton} results in unnatural limb proportions, diverging from human anatomical accuracy. See the visual details in Fig. 10 and the highest distortion in Table 2.

4.4.4 Video Models

Our experiments use the publicly available ModelScope text-to-video (Wang et al., 2023a) as the diffusion backbone. We replace this backbone with other video generation models, such as ZeroScope,² which is tuned across a range of resolutions and framerates, and showcase the frame when the keypoint reaches the end of the trajectory (i.e., the last frame) in Fig. 11. We observe that different backbones can create animations that match the text prompt, with variations in the motion patterns. Therefore, our method can be generalized to different video diffusion models, and utilizing a more powerful model can further improve the performance of our method.

4.5 Extensions

We propose two extended implementations to our system to further increase the variety and quality of animations produced.

² <https://huggingface.co/cerspense>

Table 2 Quantitative results of the ablation study with “Dyna.”, “Cons.” and “Dist.” representing Dynamism, Temporal Consistency, and Distortion, respectively. Our proposed method achieves a well-balanced trade-off among these three metrics

	Dyna. ↑	Cons. ↑	Dist. ↓
Full	20.87	0.9673	50.98
LBS	17.13	0.9785	60.56
w/o Trajectory	102.66	0.9452	65.22
Linear Interp	8.96	0.9814	42.42
X-CLIP Loss	8.16	0.9760	108.81
Image SDS	7.57	0.9779	59.07
w/o Skeleton	16.18	0.9675	265.92

The [bold] in the tables represents the best performance

4.5.1 Layered Animation

We create a single triangular mesh for clipart deformation in the clipart preprocessing stage (Sect. 3.2). However, a single mesh may struggle with self-occlusion and topological changes in animation. For example, in Fig. 12, the character's right hand overlaps with the left hand. Using a single mesh for animation cannot separate the hands to create a desirable boxing animation. To enhance our system's ability to solve these problems, we recommend shifting towards multi-layer animations by making the following modification to the pipeline:

1. Define groups with semantic meaning (e.g., a leg) and assign each shape in the SVG to a specific group. This is done manually in our experiments because no automatic grouping algorithm available currently to find meaningful movable parts.
2. The keypoint detection algorithm remains unchanged. However, since each layer occupies only a specific part

Fig. 11 Different text-to-video backbones. We display the frame when the keypoint reaches the end of the trajectory (*i.e.*, the last frame). This demonstrates that our method can be generalized to different video diffusion models

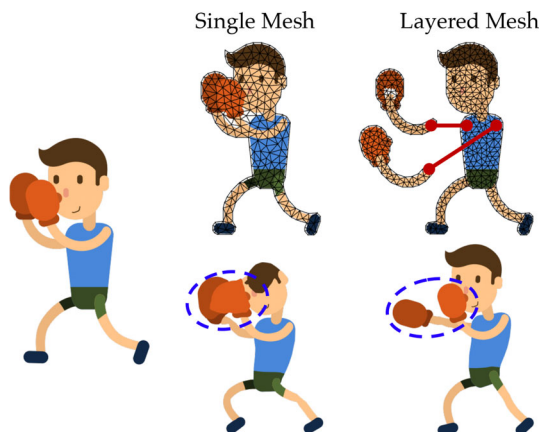
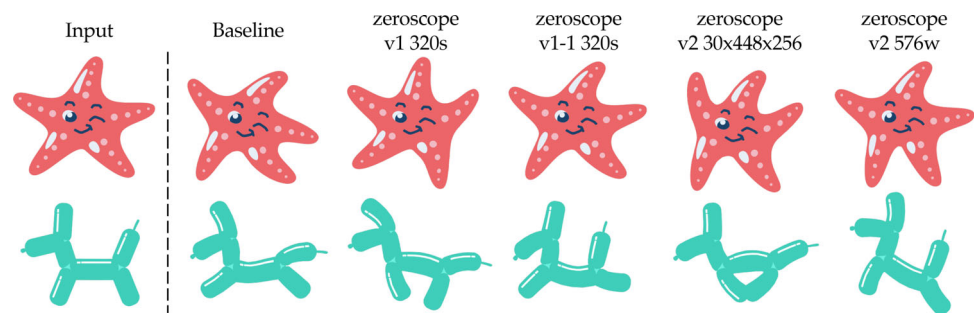


Fig. 12 Layered animation. The separation of different layers enables the animation involving topology changes. The left side shows the single layered mesh and its corresponding animation result, presenting a distorted appearance. The right side illustrates the layered mesh anchored by two keypoints, producing animations with more natural motion

of the entire object, we compute the boundary for each group and identify the keypoints within these boundaries. This ensures that each group is updated accurately based on its respective keypoints.

3. We build a triangular mesh for each layer. During optimization, each layer is deformed individually by its keypoints, and the deformed layers are then combined to form the final appearance.

We show an example in Fig. 12 to illustrate the improvement of the clipart layers and the grouping operation. Moreover, another advantage of SVG is that we do not need to reconstruct the occluded parts after grouping, as each shape in the vector graphics is already complete.

4.5.2 Higher-Order Bézier Trajectory

In our experiments, we have demonstrated that cubic Bézier curves (*i.e.*, control by four points) effectively guide the motion trajectories of keypoints to produce plausible animations. It is worth noting that the motion trajectories can be defined by higher-order Bézier curves. Introducing additional control points enables the optimization to create more

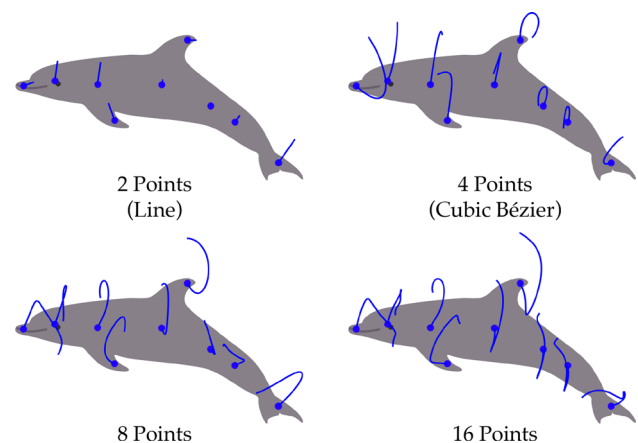


Fig. 13 The animation of optimized Bézier trajectories can be made smoother by increasing the number of control points

detailed motions. We include such examples for comparison in Fig. 13, showcasing optimized trajectories with different numbers of control points. After comparing the generated animations (see the supplementary video), we conclude that cubic Bézier curves are adequate for creating both practical and visually pleasing animations. Although higher-order curves allow for finer animation details, we need to sample more points on the curves to capture their complexity, leading to increased time and higher computation resources in creating animations. Specifically, we found that memory consumption scales linearly with the number of frames, and the time required to create each animation remains relatively stable. We leave such an “expressiveness versus complexity” trade-off to designers in actual scenarios.

5 Conclusion and Discussion

In this study, we introduce a system designed to animate clipart based on textual descriptions. Our methodology utilizes defined keypoints on the clipart and employs Bézier curves for motion trajectories, ensuring seamless and fluid movements. A key feature of our approach is its ability to generate these motion trajectories without the need for additional training datasets. This is achieved by extracting motion

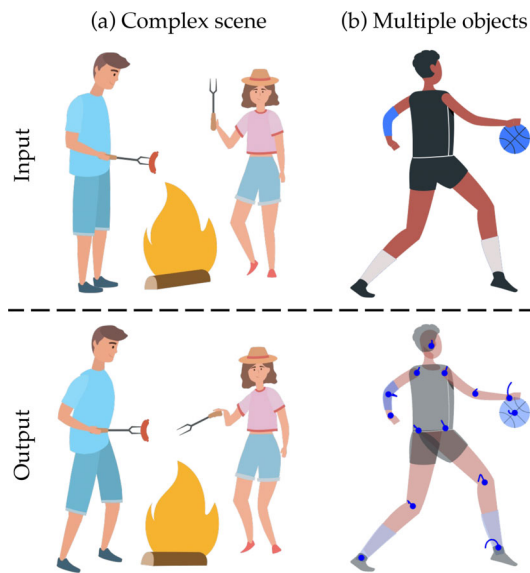


Fig. 14 Limitations Constrained by the capability of the T2V model, we observe a quality degradation when applying our method to a complex scene (a) and clipart with multiple objects (b)

priors from a pretrained text-to-video diffusion model, utilizing VSDS loss. To maintain the visual integrity of the clipart during animation, we implement constraints on keypoint trajectories through a novel skeleton loss, which are then applied to drive the clipart animation via ARAP deformation techniques. Comprehensive experiments confirm the effectiveness of our system in creating high-quality animations.

Despite the visually pleasant results produced, our AniClipart suffers from several limitations. Our system encounters difficulties when processing complex scenes and clipart with multiple objects. For instance, in Fig. 14(a), the clipart contains two characters at a picnic. Compared to applying our method to individual characters, we observe a degradation in result quality for such complex scenes. In these scenarios, character movements appear unnatural, and the resulting animation fails to correspond with the accompanying text prompt. Similarly, in Fig. 14(b), the man and the basketball are in separate layers. We expect to create an animation of the man dribbling the ball. However, the basketball's trajectory does not touch the ground, primarily due to the video diffusion model's inability to accurately generate video from the text prompt, resulting in artifacts in the distilled animation. Fortunately, the VSDS loss is not dependent on the underlying model, suggesting that a more sophisticated T2V model could help alleviate this problem.

In future research, we plan to enhance the automation of our pipeline to better support designers. Currently, our keypoint detection algorithm uses a hybrid approach that integrates both learning-based and traditional methods. To streamline this process, we could develop a specialized key-

point detection model by gathering clipart-keypoint pairs specifically for clipart images. Additionally, the current layered animation requires users to manually assign grouping information. Automating this process could be achieved by utilizing multimodal vision-language models, such as GPT-4V, to segment vector clipart into meaningful movable parts. Furthermore, we aim to expand the pipeline's functionality to handle increasingly complex tasks. By integrating 2.5D cartoon models (Rivers et al., 2010) with our system, we can simulate 3D rotation effects for 2D clipart, enhancing the complexity of animations. Lastly, we plan to explore methods for better animating complex scenes containing multiple objects, thereby broadening the real-world applicability of our approach.

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Data Availability We have included data (30 cliparts in SVG format) as electronic supplementary materials.

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